



PrintablePrograms.png

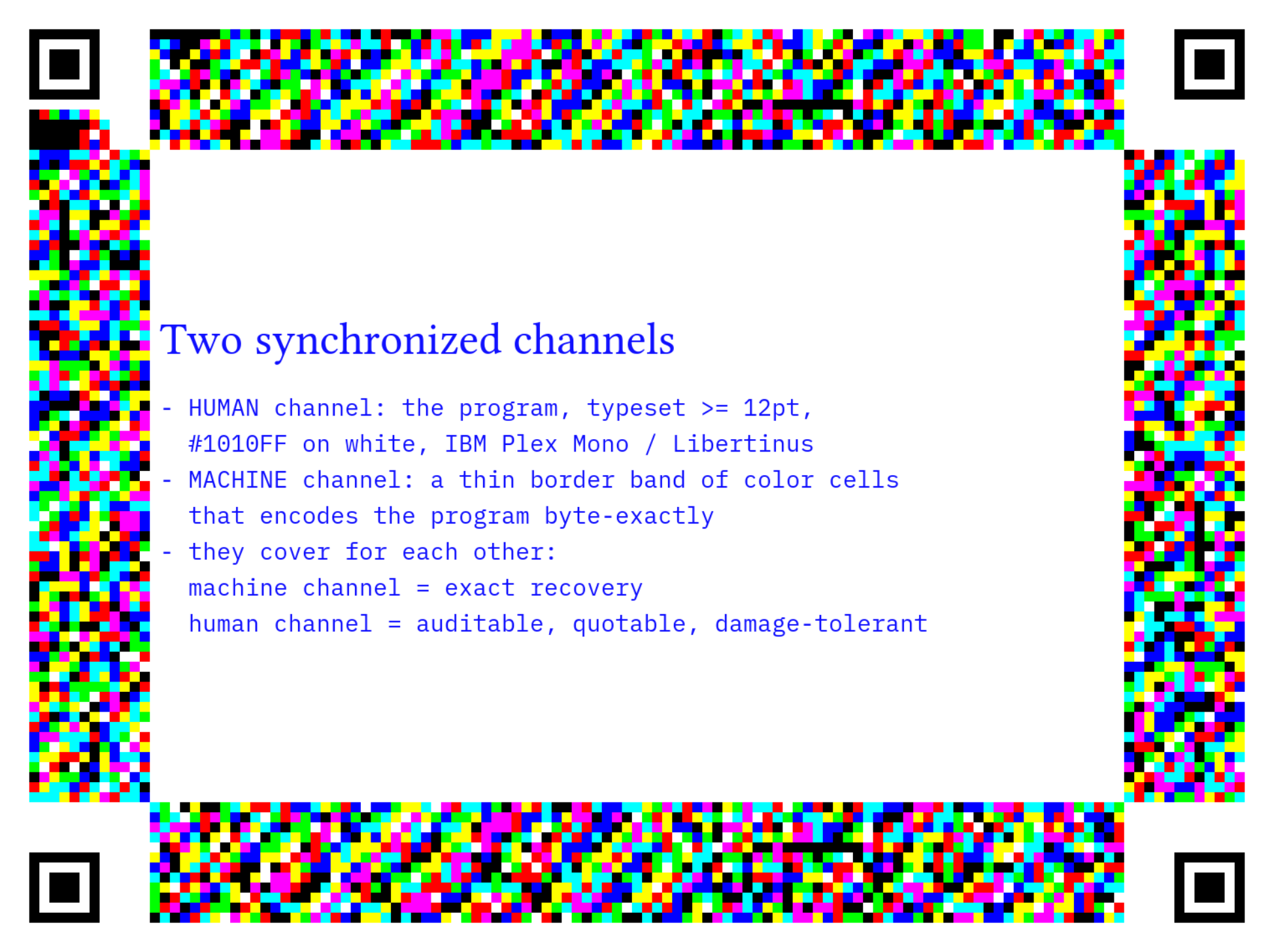
programs you can print out, pin up,
photograph, and run again

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Why paper?

- durable, human-inspectable, self-powered
- survives institutional collapse better than any disk
- magazines, punched cards, microfilm: paper was a carrier
- what paper lost when software grew: BYTE-EXACTNESS
- retyping fails; OCR fails exactly where it matters



Two synchronized channels

- HUMAN channel: the program, typeset ≥ 12 pt, #1010FF on white, IBM Plex Mono / Libertinus
- MACHINE channel: a thin border band of color cells that encodes the program byte-exactly
- they cover for each other:
machine channel = exact recovery
human channel = auditable, quotable, damage-tolerant



Anatomy of a page frame

- 8 colors at the RGB cube corners, 3 bits per cell
- QR-style finder fiducials in all four corners
- calibration strip: re-learn the palette per print
- band thickness + grid $W \times H$ self-encoded (24-bit words)
- header: hostname, IP, program id, UTC-ms filename, page + cell size in mm -> the page ENCODES ITS OWN SCALE
- CRC32-guarded packet, repeated to fill the band



The press: every format, byte-exact

`.tcl <-> .xml <-> .png <-> .jpg / .jpeg`

- canonical XML hub: base64 source + commands + real AST
- PNG tEXt chunks + JPEG COM segments carry the payload
- tcl -> jpg -> tcl through a lossy codec: byte-exact
- decoder survives rescale, blur, JPEG recompression

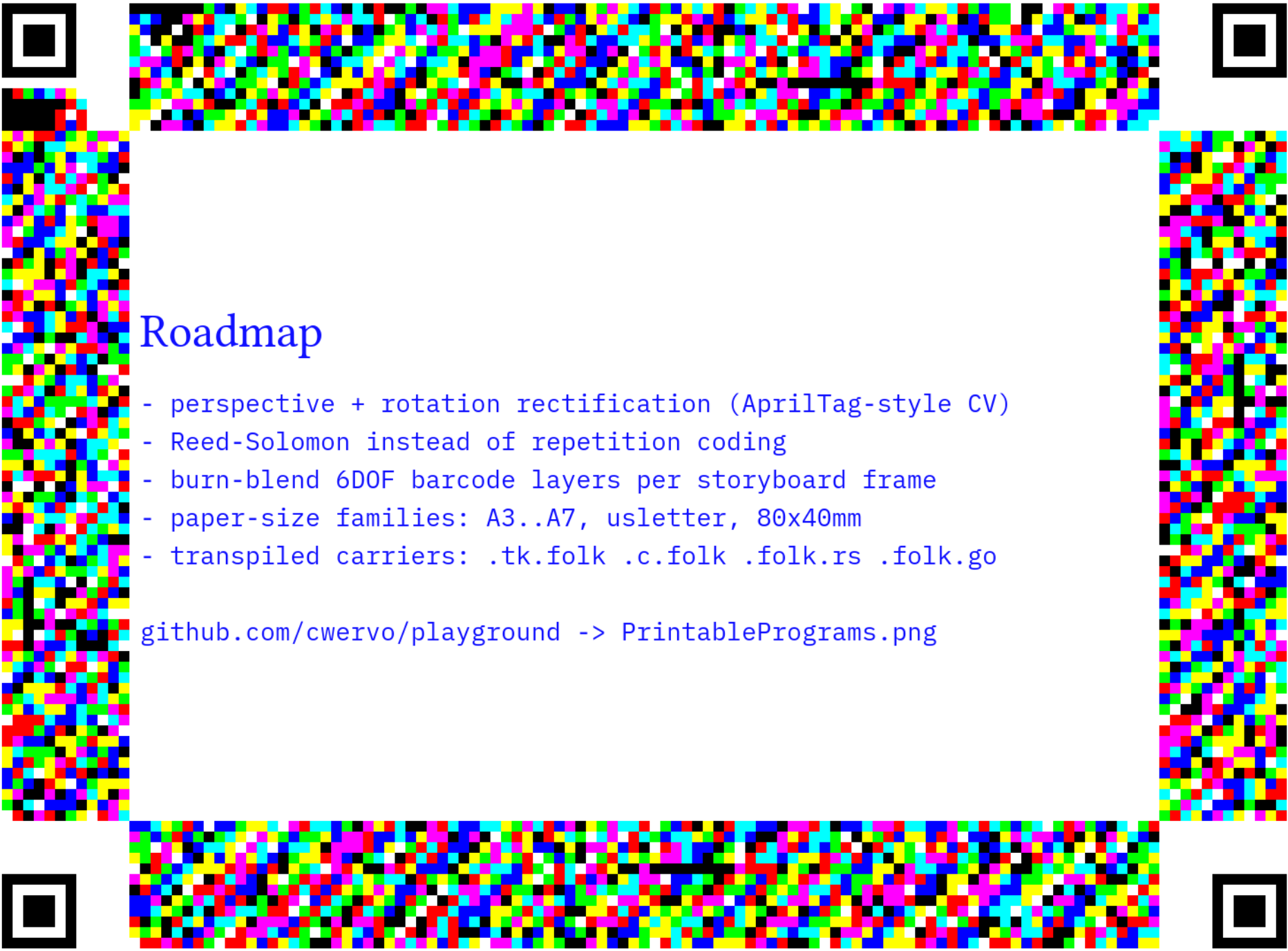
Self-demoing white paper

- WhitePaper.xml -> .md .html .pdf editions
- five US Letter page frames, ACM-typeset in Libertinus
- every printed page decodes to the paper's own source
- carriers reproduce the PDF byte-exactly when run:
 .pp.html.folk .pp.html.rust .pp.go .pp.cpp
- this talk is built the same way: each slide you are
 looking at decodes to its own text



Numbers

- 60 checks green across three suites
- ~1.7 KB per US Letter page at 2mm cells, 12-cell band
- decodes after 137% up / 63% down / blur + JPEG q80
- overflow degrades to prefix + origin URL:
<http://ip/folk-data/program/filename>



Roadmap

- perspective + rotation rectification (AprilTag-style CV)
- Reed-Solomon instead of repetition coding
- burn-blend 6DOF barcode layers per storyboard frame
- paper-size families: A3..A7, usletter, 80x40mm
- transpiled carriers: .tk.folk .c.folk .folk.rs .folk.go

github.com/cwervo/playground -> PrintablePrograms.png